

Business Rules:

1. A Person can be an Admin, a Customer, a Staff member, or any combination of them at the same time (Overlapping specialization rule O).
2. Every person in the system must provide a unique identifier, a full name, a valid email, and a phone number.
3. A Customer is the account holder who pays for reservations.
4. A Passenger is the actual traveler on the flight.
5. A Booking must be initiated by exactly one mandatory Customer, but a Customer can make many bookings over time (\$1:M\$).
6. Each Booking must have exactly one associated Payment record to track its financial status.
7. A booking can contain multiple passengers, and a passenger can appear on multiple bookings.
8. A Flight must have exactly one origin airport (DEPARTS FROM) and exactly one destination airport (ARRIVES AT).
9. An Airport is identified by a unique id.
10. An Airline operates multiple Aircraft (planes), but a single physical plane belongs to only one airline.
11. An Aircraft can be scheduled for many historical or future Flights, but a specific flight on a specific date uses exactly one aircraft.
12. A Flight must have exactly one origin airport (DEPARTS FROM) and exactly one destination airport (ARRIVES AT).
13. An Airport is identified by a standard, unique 3-character IATA code (e.g., LHE, ISB).
14. An Airline operates multiple Aircraft (planes), but a single physical plane belongs to only one airline.
15. An Aircraft can be scheduled for many flights, but a specific flight on a specific date uses exactly one aircraft.

Data Definitions / Meta Data

Table: Person (Supertype)

- **Person_ID** (INT, Primary Key)
- **Name** (VARCHAR(50))

- **Email** (VARCHAR(100))
- **Phone** (VARCHAR(15))
- **Person_Type** (char(1)) Discriminator column (e.g., 'Admin', 'Customer', 'Staff').

Table: Admin (Subtype)

- **Person_ID** (INT, Primary Key)
- **Admin_Level** (VARCHAR(15)) (e.g., 'Super', 'Staff_Manager').

Table: Customer (Subtype)

- **Person_ID** (INT, Primary Key)
- **Membership_Tier** (VARCHAR(15)) Loyalty tier (e.g., 'Gold', 'Silver', 'Regular').

Table: Staff (Subtype)

- **Person_ID** (INT, Primary Key)
- **Employee_Code** (VARCHAR(15)) — System registration code for employees.

Table: Passenger

- **Passenger_ID** (INT, Primary Key)
- **Name** (VARCHAR(50))
- **Age** (INT) (needed for child/adult safety rules).
- **Phone** (VARCHAR(15))
- **Email** (VARCHAR(100))

Table: Booking

- **Booking_ID** (INT, Primary Key)
- **Booking_Date** (DATE)

Table: Payment

- **Payment_ID** (INT, Primary Key)
- **Amount** (DECIMAL(10, 2))
- **Payment_Status** (VARCHAR(15)) (e.g., 'Success', 'Refunded').

Table: Airport

- **Airport_ID** (VARCHAR(15), Primary Key)
- **Airport_Name** (VARCHAR(50))

Table: Airline

- **Airline_ID** (INT, Primary Key)
- **Airline_Name** (VARCHAR(50))

Table: Aircraft

- **Aircraft_Tail_ID** (VARCHAR(15), Primary Key)
- **Model** (VARCHAR(30))

Table: Flight

- **Flight_Number** (VARCHAR(15), Primary Key)
- **Departure_Time** (TIMESTAMP)
- **Arrival_Time** (TIMESTAMP)

Table: Seats

- **Seat_ID** (INT, Primary Key)

3. Weak Entities & Sub-Components

Table: Ticket

- **Ticket_ID** (INT, Primary Key)
- **Price** (DECIMAL(10,2))

Table: Baggage (Weak Entity)

- **Baggage_ID** (INT, Primary Key)
- **Bags_Count** (INT)
- **Type** (VARCHAR(25)) (e.g., 'Checked-In', 'Carry-On').

Junction / Associative Entities (Base Form)

Table: Booking_Flyers

- **Flyer_Status** (VARCHAR(20)) — (e.g., 'Checked-In', 'No-Show', 'Boarded').

Table: Booking_Flight

- **Layover_Duration** (INT) — Total waiting time in minutes during connections.
- **Flight_Sequence** (INT) — The order of the flight leg (e.g., 1 for the first flight, 2 for the layover flight).

Table: Book_Seats

- **Class** (VARCHAR(15))
- **Seat Number** (VARCHAR(10))

ER Diagram

